

Trainings ▾

Preparatory Steps

Automotive With CM871

- If necessary, remove the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-370.html>)



Remove

Automotive with CM870

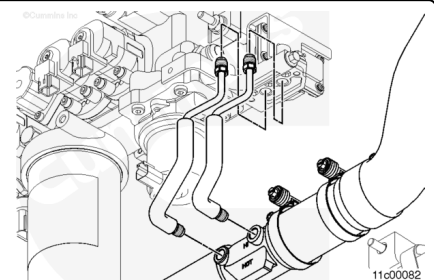
For engines beginning with ESN First 14040552, the pressure sensor does **not** have to be removed to allow the removal of the EGR differential pressure sensor tubes.

Pressure sensor removal is necessary **only** if the sensor adapter is being replaced. Refer to Procedure 011-028 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-028.html>)

Tag the ends of the tubes before removing the sensor tubes. Incorrectly installing the tubes to the sensor adapter will result in a fault code.

Loosen the tube nuts at the front EGR connection tube and the sensor adapter.

Remove the tubes.



Automotive With CM871

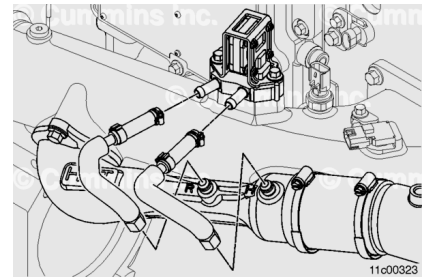
Tag the ends of the tubes before removing the sensor tubes.

Incorrectly installing the tubes to the sensor adapter will result in a fault code.

Loosen the tube nuts at the EGR venturi tube.

Loosen the hose clamps at the sensor adapter.

Remove the tubes.

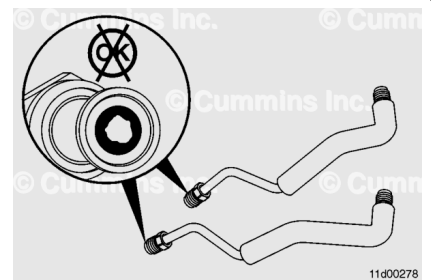


Clean and Inspect for Reuse

Automotive with CM870

Carbon deposits inside the tubes can cause an amber CHECK ENGINE light, in addition to any combination of the following:

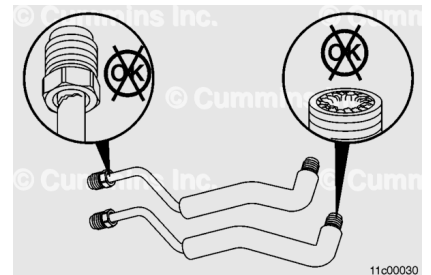
- Low power
- Black smoke
- Engine surge
- Engine runs rough
- Exhaust gas recirculation valve delta pressure fault code.



Note : If the exhaust gas recirculation (EGR) cooler is being replaced due to an internal coolant leak, the EGR differential pressure tubes must be replaced.

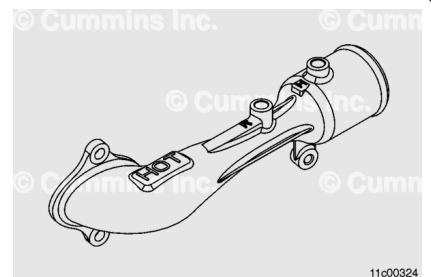
Check the inside of the tubes.

If a tube is clogged or partially clogged, replace the tube.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the mating orifices for the EGR crossover tubes.

If clogged, saturate the carbon deposit with a mineral-based solvent, or equivalent.

Carefully clean debris from the mouth of the tube, making sure **not** to damage the orifice.

Dry with compressed air.

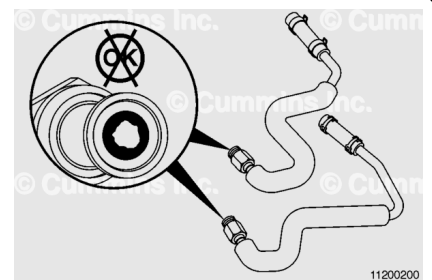
Refer to Procedure 011-022 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)

Automotive With CM871

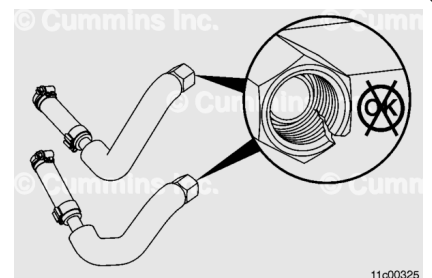
Carbon deposits inside the tubes can cause an amber CHECK ENGINE light, in addition to any combination of the following:

- Low power
- Black smoke
- Engine surge
- Engine runs rough
- Exhaust gas recirculation valve differential pressure fault code
- Progressive damage to the Cummins® particulate filter.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : If coolant is found in the EGR differential pressure tubes, they must be replaced.

Check the inside of the tubes for carbon deposits.

If a tube is partially clogged, saturate the carbon deposits inside of the tube with mineral based solvent, or equivalent.

Carefully clean the debris from the mouth of the tube. Make sure **not** to damage the tube.

If the tube is fully clogged or otherwise unable to be cleaned, the tube **must** be replaced.

Clean the tube with safety solvent.

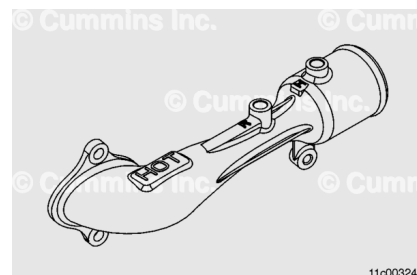
Dry with compressed air.

Check the tubes, hose, and hose clamps for cracks or other damage.

Replace the tubes if the tubes or hose is damaged. Replace the hose clamps if damaged.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the mating orifices of the EGR crossover tubes.

If clogged, saturate the carbon deposit with a mineral-based solvent, or equivalent.

Carefully clean the debris from the mouth of the orifice, make sure **not** to damage the orifice.

Dry with compressed air.

Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)

Install

Automotive with CM870

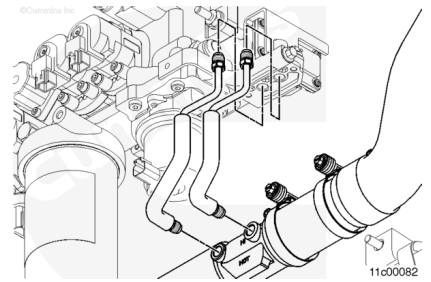
For engines beginning with ESN 14040552, the tubes **must** be installed so they do **not** cross each other.

Apply anti-seize compound to the tube threads and install the tubes.

Torque Value: 17 n·m [150 in-lb]

Position the insulation on the differential pressure sensor tubes so the tube nuts at the venturi connections are covered.

This insulation is designed to **only** cover the bottom portion of the tube which threads to the EGR tube. This location proves to be most effective and prevents heat soaking of the EGR differential pressure sensor.



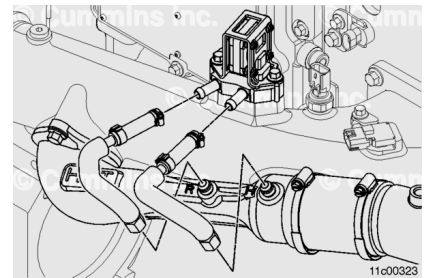
Automotive With CM871

The tubes **must** be installed so they do **not** cross each other.

Apply anti-seize compound to the tube threads and install the tubes.

Tighten the tube nuts at the EGR venturi connection.

Torque Value: 26 n·m [230 in-lb]

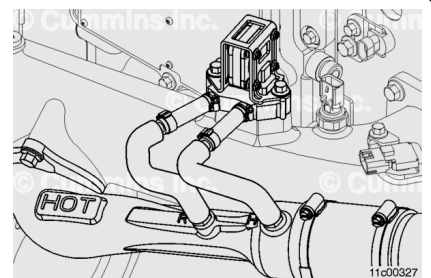


Tighten the hose clamps at the differential pressure sensor adapter.

Torque Value: 1.5 n·m [13 in-lb]

Position the insulation on the differential pressure sensor tubes so the tube nuts at the venturi connection are covered.

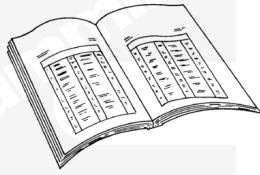
The insulation on the pressure tubes is designed to **only** cover the bottom portion of the tube which threads into the EGR transfer tube. This is the most effective location and also reduces the possibility of heat soaking of the EGR differential pressure sensor.



Finishing Steps

Automotive with CM870

- If removed, install the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/70/70-019-370.html>)
- Start the engine and verify proper operation.
- Check for fault codes.



Automotive With CM871

- If removed, install the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)
- Start the engine and verify proper operation.
- Check for fault codes.
- If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)

